

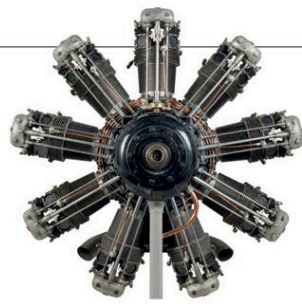
ENGINE COOLING

AIR-COOLED

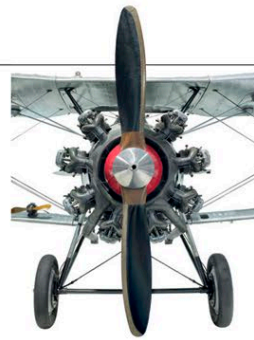
Cooling an engine with the air stream requires controlled distribution of flow, which favors the radial layout, where the cylinders are all equally “exposed to the breeze.” Most light aircraft today have air-cooled engines.

WATER-COOLED

Cooling the engine by getting liquid coolant to flow around the critical, hottest parts of an engine is easier than air-cooling, and allows high power output and efficiency. However, the radiator(s) cause aerodynamic drag and are vulnerable to damage.



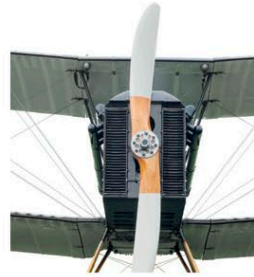
BRISTOL JUPITER ENGINE



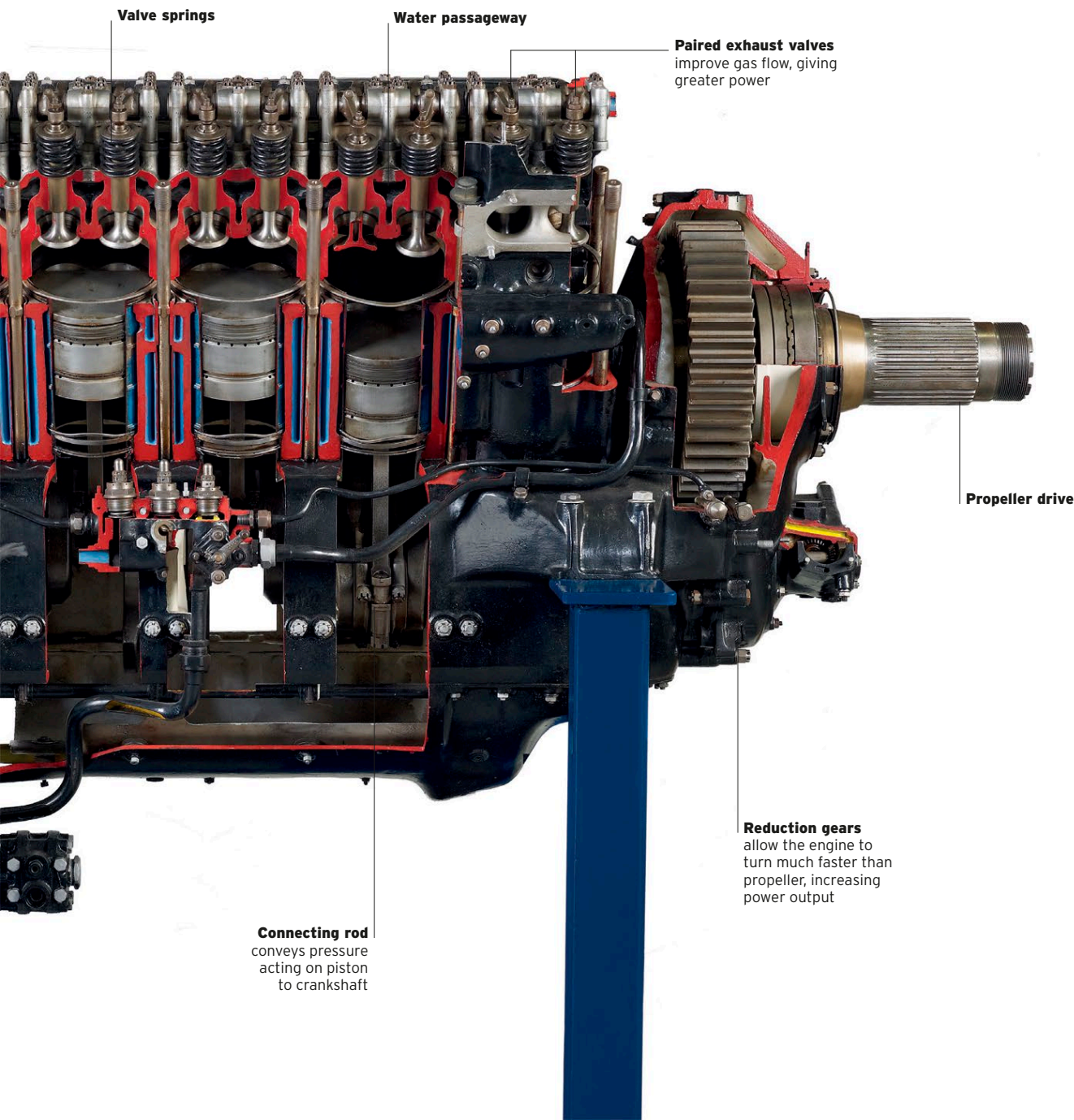
BRISTOL BULLDOG FIGHTER



HISPANO-SUIZA V8 ENGINE



ROYAL AIRCRAFT FACTORY S.E.5A

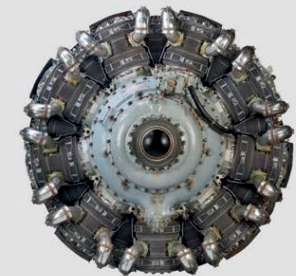


ENGINE CONFIGURATIONS

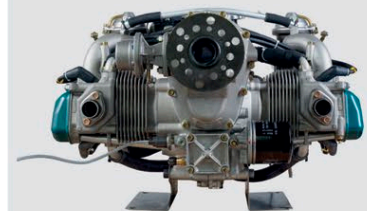
While many different engine layouts have been tried, from Anzani's simple three-cylinder “fan” of 1908 to the 24-cylinder, H-configuration Napier Sabre of the 1940s, three basic types have come to the fore.

**V TYPE**

The inline engine has minimal frontal area; however, as more and more cylinders are added, it becomes overlong and tends to bend in the middle. Combining two “inlines” on one crankshaft makes the compact and strong V.

**RADIAL**

Arranging the “pots” around a single-throw crank produces a very compact unit that is ideal for air-cooling—the radial or “R.” Extra banks can be added, giving more power for a given diameter, but making cooling a challenge.

**OPPOSED**

US manufacturer Continental made the “O” type of engine popular for light aircraft use. Lycoming and other makers quickly followed suit. The opposed, or flat, engine is compact, lends itself to air-cooling, and allows a good view over the nose.